

Date	8 July 2026	Project Name	Heart Disease Analysis
Team ID	—	Maximum Marks	5 Marks

PROJECT PLANNING PHASE

Project Planning Template (Product Backlog, Sprint Planning & Agile Estimations)

Team Leadership & Roles: Mounika Sanapathi (Team Lead) | Thakasi Balaji | Kale Sai Praneeth | Navyasri Gonnuri | Shanthi Kumari Challa

Product Backlog, Sprint Schedule, and Estimation

The table below presents the structured product backlog, distributed over four key sprint iterations, outlining precise user stories, estimates, priorities, and assignments.

Sprint	Epic / Requirement	User Story	User Story / Task	Points	Priority	Assigned Member
Sprint-1	Data Collection	USN-1	Collect clinical cardiovascular dataset.	3	HIGH	Mounika Sanapathi
Sprint-1		USN-2	Clean dataset & resolve schema errors.	5	HIGH	Thakasi Balaji
Sprint-1		USN-3	Prepare dataset & scale physiological metrics.	5	HIGH	Kale Sai Praneeth
Sprint-1		USN-4	Validate data integrity against medical standards.	3	MEDIUM	Navyasri Gonnuri
Sprint-2	Tableau Dashboard	USN-5	Create core KPI clinical overview dashboard.	5	HIGH	Shanthi Kumari Challa
Sprint-2		USN-6	Chest pain type & distribution charts.	3	HIGH	Thakasi Balaji
Sprint-2		USN-7	Physiological metric analysis (Cholesterol/BP).	5	HIGH	Kale Sai Praneeth
Sprint-2		USN-8	ECG metrics & angina risk tier tracking.	3	MEDIUM	Mounika Sanapathi
Sprint-3	Insights	USN-9	Demographic profile analysis (Age brackets).	3	HIGH	Navyasri Gonnuri
Sprint-3		USN-10	Max heart rate achieved velocity distribution.	5	HIGH	

Sprint	Epic / Requirement	User Story	User Story / Task	Points	Priority	Assigned Member
						Shanthi Kumari Challa
Sprint-3		USN-11	ST depression severity condition patterns.	3	MEDIUM	Thakasi Balaji
Sprint-3		USN-12	Hospital branch & regional volume analysis.	3	HIGH	Mounika Sanapathi
Sprint-3		USN-13	Implement interactive cross-filtering views.	2	MEDIUM	Navyasri Gonnuri
Sprint-4	Deployment	USN-14	Develop Flask web application wrapper interface.	5	HIGH	Navyasri Gonnuri
Sprint-4		USN-15	Assemble sequential Tableau Story presentation.	3	HIGH	Mounika Sanapathi
Sprint-4		USN-16	GitHub web deployment & code synchronization.	3	MEDIUM	Kale Sai Praneeth
Sprint-4		USN-17	Conduct end-to-end performance & visual testing.	3	HIGH	Shanthi Kumari Challa
Sprint-4		USN-18	Compile technical clinical project documentation.	2	HIGH	Shanthi Kumari Challa

Project Tracker, Velocity & Burndown Chart

Sprint Iteration	Total Story Points	Duration (Planned)	Sprint Start Date	Sprint End Date	Story Points Completed	Actual Release Date
Sprint-1	16	7 Days	03 Aug 2026	09 Aug 2026	16	09 Aug 2026
Sprint-2	16	7 Days	10 Aug 2026	16 Aug 2026	16	16 Aug 2026
Sprint-3	16	7 Days	17 Aug 2026	23 Aug 2026	16	23 Aug 2026
Sprint-4	13	7 Days	24 Aug 2026	30 Aug 2026	13	30 Aug 2026

Velocity Calculations

A burn down chart visualizes work left to do versus time. Team performance values are calculated below:

Calculated Team Velocity: $61 / 4 = 15.25$ story points completed per sprint tier iteration.

For context, if a 10-day sprint is assumed with a theoretical velocity of 20, the average velocity iteration unit equates to:

$$AV = \Sigma(\text{Story Points}) / \Sigma(\text{Sprint Days}) = 20 / 10 = 2$$

Sprint Burndown Chart (Work Remaining vs Iterations)

